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Roll No

CE-5005(1)-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Water Resources Engineering

Time : Three Hours

Maximum Marks : 70

Note: i) Answer any five questions.

ii) All questions carry equal marks.

iii) Draw flow charts and diagrams, where needed.

1. a) Explain the terms 'duty' and 'delta'. Drive a relationship between the two. 7
b) A water course has culturable commanded area of 2600 hectares, out of which the intensities of irrigation for perennial sugar-cane and rice crops are 20% and 40% respectively. The duty for these crops at the head of water course are 750 hectares/cumec and 1800 hectares/cumec respectively. Find the discharge required at the head of water course if the peak demand is 20% of the average requirement. 7
2. a) What are the advantages and disadvantages of well irrigation over canal irrigation? 7
b) Enlist the sources of underground water. Discuss infiltration galleries. 7
3. a) Describe the various methods of computing average rainfall over a basin. Discuss the merits and demerits of each one. 7
b) What are the assumptions made in unit hydrograph theory? Discuss the limitations of unit hydrograph theory. 7

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4. a) Distinguish between a contour canal and a ridge canal. 7
b) Design a regime channel for a discharge of $35 \text{ m}^3/\text{sec}$. with silt factor of 0.9 by Lacey's theory, taking side slopes as 1H:2V. 7
5. a) What is Probable Maximum Flood (PMF), Standard Project Flood (SPF) and flood of a specific return period. 7
b) Write a short note on flood control measures. 7
6. a) What are the various types of irrigation efficiencies? Discuss. 7
b) Enlist the various types of spillway used and indicate the situation that warrant their suitability and choice. 7
7. The isohyets drawn for a storm which occurred over a drainage basin area 950 km^2 yielded the following information. 14

Isohyets interval (in mm)	85-75	75-65	65-55	55-45	45-35
Area between isohytes (km^2)	125	236	264	175	150

Determine the average depth of rainfall over the basin and the volume of runoff generated, if there were no losses.

8. Write short notes on: 14
- i) Infiltration index
 - ii) Flood frequency
 - iii) Evaporation and Transpiration
 - iv) Crop rotation
